Evaluation runs: 2pc-tr (two-pc-trillium)

generated 2026-11-03 from brain-data 03b0b57cc 2026-11-03 11:21:51 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): . Blue rows are at the current pins (TraceForge 9a0629e, merge join 367cdc6, JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	4	12	conforms	0.0	4	1 h, 250 GB	2	4	brain03, 2026-11-02
3	8	87	conforms	0.0	4	1 h, 250 GB	2	9	brain03, 2026-11-02
4	16	888	conforms	0.1	4	1 h, 250 GB	2	17	brain03, 2026-11-02
5	32	16365	conforms	0.4	4	1 h, 250 GB	2	27	brain03, 2026-11-02
6	64	531648	conforms	8.4	4	1 h, 250 GB	2	20	brain03, 2026-11-02
7	128	25499439	conforms	222.8	46	1 h, 250 GB	2	57	brain03, 2026-11-02
8			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

stateless: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2	4	10	4	violation	0.0	4	1 h, 250 GB	6	19	brain03, 2026-11-02
3	8	96	54	violation	0.0	4	1 h, 250 GB	56	435	brain03, 2026-11-02
4	16	1560	960	violation	0.8	4	1 h, 250 GB	962	15371	brain03, 2026-11-02
5	32	39120	24600	violation	13.9	4	1 h, 250 GB	24602	787206	brain03, 2026-11-02
6	64	1409040	889920	violation	428.9	16	1 h, 250 GB	889922	56954928	brain03, 2026-11-02
7	128			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-02

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 7 ignore-abort-vote bag, 7 bag ignore-abort-vote, 7 bag correct, 8 correct bag, 8 bag ignore-abort-vote, 8 bag correct, 9 bag ignore-abort-vote, 9 bag correct, 10 bag correct, 10 bag ignore-abort-vote, 11 bag ignore-abort-vote, 11 bag correct.

skipped: a smaller size had reached its limit: 8 ignore-abort-vote bag, 9 correct bag, 9 ignore-abort-vote bag, 10 correct bag, 10 ignore-abort-vote bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	4	12	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	8	87	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4	16	888	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-02
5	32	16365	conforms	0.5	4	1 h, 250 GB	brain02, 2026-11-02
6	64	531648	conforms	9.5	2	1 h, 250 GB	brain02, 2026-11-02
7	128	25499439	conforms	866.8	46	48 h, 250 GB	brain05, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
8			(1 h)			1 h, 250 GB	brain02, 2026-11-02

stateful: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	4	10	4	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3	8	96	54	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4	16	1560	960	violation	0.2	4	1 h, 250 GB	brain02, 2026-11-02
5	32	39120	24600	violation	1.6	4	1 h, 250 GB	brain02, 2026-11-02
6	64	1409040	889920	violation	21.7	8	1 h, 250 GB	brain02, 2026-11-02
7	128	69048000	43641360	violation	1201.2	132	48 h, 250 GB	brain05, 2026-11-02
8				(1 h)			1 h, 250 GB	brain02, 2026-11-02

Not run (stateful). left out by the table rules (ok): 7 ignore-abort-vote bag, 7 correct bag.

skipped: a smaller size had reached its limit: 9 correct bag, 9 ignore-abort-vote bag, 10 correct bag, 10 ignore-abort-vote bag.

planned but never scheduled (grid points beyond the ladder): 8 bag ignore-abort-vote, 8 bag correct, 9 bag ignore-abort-vote, 9 bag correct, 10 bag correct, 10 bag ignore-abort-vote, 11 bag ignore-abort-vote, 11 bag correct.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12	conforms	0.1	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	8	87	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
4	16	888	conforms	0.2	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
5	32	16365	conforms	0.9	4	10 min, 16 GB	0.0	0.8	0.0	in-memory	brain05, 2026-11-02
6	64	531648	conforms	10.9	24	1 h, 250 GB	0.0	10.9	0.0	merge	brain05, 2026-11-03
7	128	25499439	conforms	318.9	42	1 h, 250 GB	0.1	318.9	0.0	merge	brain05, 2026-11-03

offline: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	10	2	violation	0.1	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	8	96	6	violation	0.2	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
4	16	1560	14	violation	0.3	4	10 min, 16 GB	0.0	0.3	0.0	in-memory	brain05, 2026-11-02
5	32	39120	30	violation	2.1	6	1 h, 250 GB	0.0	2.0	0.1	merge	brain05, 2026-11-03
6	64	1409040	62	violation	29.4	26	1 h, 250 GB	0.0	29.3	0.0	merge	brain05, 2026-11-03
7	128	69048000	126	violation	1114.3	118	1 h, 250 GB	0.1	1114.3	0.0	merge	brain05, 2026-11-03

Not run (offline). skipped: a smaller size had reached its limit: 8 ignore-abort-vote bag, 8 correct bag, 9 correct bag, 9 ignore-abort-vote bag, 10 correct bag, 10 ignore-abort-vote bag.

planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 8 bag ignore-abort-vote, 8 bag correct, 9 bag ignore-abort-vote, 9 bag correct.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	12	conforms	2.8	160	10 min, 16 GB	2.7	0.0	0.0	in-memory	brain01, 2026-11-02
3		87	conforms	0.3	4	10 min, 250 GB		0.1	0.3	in-memory	brain01, 2026-11-03
4		888	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	4	10	2	violation	2.7	160	10 min, 16 GB	2.7	0.0	0.0	in-memory	brain01, 2026-11-02
3		96	6	violation	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
4		1560	14	violation	0.0	4	10 min, 250 GB		0.3	0.0	in-memory	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote.jmcspec-tfimpl, 7 bag correct.jmcspec-tfimpl, 8 bag correct.jmcspec-tfimpl, 8 bag ignore-abort-vote.jmcspec-tfimpl, 9 bag ignore-abort-vote.jmcspec-tfimpl, 9 bag correct.jmcspec-tfimpl.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	4	12	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	8	87	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
4	16	888	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	4	10	4	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	8	96	54	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
4	16	1560	960	violation	0.2	4	1 h, 250 GB	2	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote, 7 bag correct, 8 bag ignore-abort-vote, 8 bag correct, 9 bag ignore-abort-vote, 9 bag correct.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.1	153	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3		87	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
4		888	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **ignore-abort-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	10	10	conforms	3.2	172	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3		96	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
4		1560	conforms	0.1	4	10 min, 250 GB		0.3	0.1	in-memory	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	10	2	violation	3.1	153	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3		96	6	violation	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
4		1560	14	violation	0.1	4	10 min, 250 GB		0.3	0.1	in-memory	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). planned but never scheduled (grid points beyond the ladder): 7 bag correct.jimpl-tfimpl, 7 bag ignore-abort-vote.jimpl-tfimpl, 7 bag ignore-abort-vote.vs-correct.jimpl-tfimpl, 8 bag ignore-abort-vote.vs-correct.jimpl-tfimpl, 8 bag ignore-abort-vote.jimpl-tfimpl, 8 bag correct.jimpl-tfimpl.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.1	153	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	87		conforms	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
4	888		conforms	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **ignore-abort-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	10	10	conforms	3.1	172	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	96		conforms	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
4	1560		conforms	0.0	4	10 min, 250 GB	0.3		0.0	in-memory	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	10	2	violation	3.1	172	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	87		6	violation	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
4	888		14	violation	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote.vs-correct.tfimpl-jimpl, 7 bag ignore-abort-vote.tfimpl-jimpl, 7 bag correct.tfimpl-jimpl, 8 bag ignore-abort-vote.vs-correct.tfimpl-jimpl, 8 bag correct.tfimpl-jimpl, 8 bag ignore-abort-vote.tfimpl-jimpl.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.2	164	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3		87	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03
4		888	conforms	0.1	4	10 min, 250 GB		0.1	0.1	in-memory	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **ignore-abort-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	10	10	conforms	3.1	160	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3		96	conforms	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
4		1560	conforms	0.1	4	10 min, 250 GB		0.3	0.1	in-memory	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	10	2	violation	3.2	164	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
3		96	6	violation	0.0	4	10 min, 250 GB		0.1	0.0	in-memory	brain01, 2026-11-03
4		1560	14	violation	0.0	4	10 min, 250 GB		0.3	0.0	in-memory	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). planned but never scheduled (grid points beyond the ladder): 7 bag correct.jimpl-tfimpl, 7 bag ignore-abort-vote.jimpl-tfimpl, 7 bag ignore-abort-vote.vs-correct.jimpl-tfimpl, 8 bag ignore-abort-vote.vs-correct.jimpl-tfimpl, 8 bag ignore-abort-vote.jimpl-tfimpl, 8 bag correct.jimpl-tfimpl.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	12	conforms	3.2	164	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	87		conforms	0.2	4	10 min, 250 GB	0.1		0.2	in-memory	brain01, 2026-11-03
4	888		conforms	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **ignore-abort-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	10	10	conforms	3.1	160	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	96		conforms	0.1	4	10 min, 250 GB	0.1		0.1	in-memory	brain01, 2026-11-03
4	1560		conforms	0.0	4	10 min, 250 GB	0.3		0.0	in-memory	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	12	10	2	violation	3.1	160	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
3	87		6	violation	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03
4	888		14	violation	0.0	4	10 min, 250 GB	0.1		0.0	in-memory	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). planned but never scheduled (grid points beyond the ladder): 7 bag ignore-abort-vote.vs-correct.tfimpl-jimpl, 7 bag ignore-abort-vote.tfimpl-jimpl, 7 bag correct.tfimpl-jimpl, 8 bag ignore-abort-vote.vs-correct.tfimpl-jimpl, 8 bag correct.tfimpl-jimpl, 8 bag ignore-abort-vote.tfimpl-jimpl.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12	12	conforms	3.1	160	1 h, 250 GB		brain01, 2026-11-02
3	87	87	conforms	8.2	162	1 h, 250 GB		brain01, 2026-11-03
4	888	888	conforms	13.2	2044	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **ignore-abort-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	10	10	conforms	3.1	158	1 h, 250 GB		brain01, 2026-11-02
3	96	96	conforms	8.1	170	1 h, 250 GB		brain01, 2026-11-03
4	1560	1560	conforms	16.4	2129	1 h, 250 GB		brain01, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **ignore-abort-vote**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	12	10	4	violation	3.1	160	1 h, 250 GB	1881	brain01, 2026-11-02
3	87	96	54	violation	7.9	165	1 h, 250 GB	4054	brain01, 2026-11-03
4	888	1560	960	violation	16.9	2116	1 h, 250 GB	4494	brain01, 2026-11-03

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 7 bag correct.w16, 7 bag ignore-abort-vote.w16, 7 bag ignore-abort-vote.vs-correct.w16, 8 bag correct.w16, 8 bag ignore-abort-vote.vs-correct.w16, 8 bag ignore-abort-vote.w16.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	12	12	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	87	87	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
4	888	888	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **ignore-abort-vote**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	10	10	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	96	96	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
4	1560	1560	conforms	0.1	4	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair ignore-abort-vote, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	12	10	4	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	87	96	54	violation	0.0	4	1 h, 250 GB	6	brain01, 2026-11-02
4	888	1560	960	violation	0.2	4	1 h, 250 GB	2	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). planned but never scheduled (grid points beyond the ladder): 7 bag correct, 7 bag ignore-abort-vote, 8 bag correct, 8 bag ignore-abort-vote.